Room Acoustics of a Residential Apartment

Modal Analysis, Sound Propagation, and Acoustic Treatment

COMSOL Multiphysics Project Report

Acoustics and Vibrations

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SD2175 — Course Project

May 30, 2026

Abstract

This report investigates the acoustic behaviour of a residential apartment using a two-dimensional finite element model built in COMSOL Multiphysics. The study employs the Pressure Acoustics, Frequency Domain interface to compute eigenfrequencies, mode shapes, and sound pressure level (SPL) distributions in the frequency range 20 Hz to 500 Hz. Two loudspeakers placed in the living room serve as the primary internal source, while an external road-noise source above the apartment is modelled using a Normal Acceleration BC on a dedicated exterior air domain, with sound entering through both the top wall and a balcony door opening. The influence of interior door states — fully open, closed, and with a realistic 2 mm to 5 mm gap — on the coupled room modes is examined through parametric studies. Absorbing boundary conditions are used to evaluate the effectiveness of passive bass-trap treatment at modal pressure antinodes. Results quantify the frequency shift induced by door closure, the leakage through sub-millimetre gaps, and the degree of SPL reduction achievable with corner absorbers, leading to practical recommendations on speaker placement and room treatment.

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1 Introduction

1.1 Motivation

The acoustic quality of a listening room is strongly influenced by the geometry and boundary conditions of the enclosed space. When loudspeakers reproduce music or film audio in a domestic room, the sound field is not simply the direct radiation from the drivers: it is the superposition of the direct field and a complex pattern of reflections that builds up standing waves at specific frequencies. At these *resonance frequencies* — commonly called *room modes* — the sound pressure varies dramatically from one point in the room to another, producing severe colouration of the perceived sound.

A particular challenge in apartments is that the acoustic volume is not a simple rectangular enclosure. Interior doors, connecting hallways, and irregularly shaped floor plans all modify the set of resonance frequencies and the spatial distribution of standing waves. The door state — open or closed — changes the coupled air volume and therefore the modal structure, while even a small gap under a closed door provides a significant low-frequency acoustic short-circuit between adjacent rooms.

In addition to the internally generated sound from loudspeakers, apartments are exposed to external noise sources such as road traffic. This external excitation arrives as a spatially distributed pressure field impinging on the building envelope, coupling into the room through the finite transmission loss of the walls, and potentially exciting the same room modes as the internal speakers, but with a different spatial weighting.

1.2 Objectives

The goals of this project are:

(i) To build a geometrically faithful 2D COMSOL model of a real residential apartment from measured floor-plan coordinates.

(ii) To identify the dominant eigenfrequencies and mode shapes of the entire apartment in the range 20 Hz to 500 Hz, with particular focus on the living room as the primary listening space and on how sound propagates into adjacent rooms.

(iii) To quantify how closing the living-room door shifts the modal frequencies and affects the SPL distribution throughout the apartment.

(iv) To model door-gap leakage using a parametric sweep of gap heights $\{1, 2, 3, 4, 5\}$ mm implemented as a shrinking door leaf rectangle, and assess the acoustic impact on both the source room and adjacent bedroom.

(v) To compare the sound field excited by internal loudspeakers with that driven by an external road-noise source modelled above the apartment.

(vi) To evaluate passive acoustic treatment (corner absorbers) and formulate practical recommendations for improving the listening experience.

1.3 Scope and assumptions

The model is limited to a 2D top-down (horizontal) cross-section of the apartment. This is justified by the flat, uniform ceiling, which decouples vertical modes from horizontal modes to a good approximation in the frequency range of interest. Furniture is excluded from the base model; its effect is a secondary scattering contribution that becomes relevant only above 500 Hz, outside the scope of this study. All simulations use air at standard conditions ($c = 343\,\text{m/s}$, $\rho_0 = 1.21\,\text{kg/m}^3$).

2 Physical Background

2.1 The wave equation and the Helmholtz equation

Sound propagation in a quiescent, lossless, homogeneous fluid is governed by the linear acoustic wave equation[1,2],

$$\nabla^2 p - \frac{1}{c^2}\frac{\partial^2 p}{\partial t^2} = q_s, \tag{1}$$

where $p(\mathbf{x}, t)$ is the acoustic pressure, c the speed of sound, and q_s a source term. For time-harmonic fields of the form $p(\mathbf{x}, t) = \text{Re}[\hat{p}(\mathbf{x})\, e^{i\omega t}]$, eq. (1) reduces to the *Helmholtz equation*,

$$\nabla^2 \hat{p} + k^2 \hat{p} = -i\omega \hat{q}_s, \qquad k = \frac{\omega}{c} = \frac{2\pi f}{c}, \tag{2}$$

where k is the wavenumber and $\hat{p}$ is the complex pressure amplitude. Equation (2) is the equation solved by the COMSOL *Pressure Acoustics, Frequency Domain* interface[3].

2.2 Room modes and eigenfrequencies

When eq. (2) is solved subject to homogeneous Neumann boundary conditions ($\partial\hat{p}/\partial n = 0$ on all walls, corresponding to perfectly rigid boundaries), only specific discrete frequencies — the *eigenfrequencies* f_n — admit non-trivial solutions. These solutions are the *mode shapes* $\hat{p}_n(\mathbf{x})$, which describe the spatial distribution of pressure for each resonance.

For the idealised case of a rectangular room of dimensions $L \times W$, the eigenfrequencies are given analytically by[2]

$$f_{n,m} = \frac{c}{2}\sqrt{\left(\frac{n}{L}\right)^2 + \left(\frac{m}{W}\right)^2}, \qquad n, m = 0, 1, 2, \ldots \tag{3}$$

Modes with only one non-zero index are *axial* (one pair of parallel walls), those with two non-zero indices are *tangential*, and those with three are *oblique*. Because the apartment under study has a non-rectangular, irregular floor plan, eq. (3) does not apply directly; the eigenfrequencies must be computed numerically by COMSOL's eigenfrequency solver. Equation (3) is nonetheless useful for an order-of-magnitude check of the first few modes.

2.3 Boundary conditions

2.3.1 Sound-hard (rigid) wall

A rigid wall imposes zero normal particle velocity at the surface,

$$\frac{\partial \hat{p}}{\partial n} = 0. \tag{4}$$

This is the default condition applied to all structural boundaries in the baseline model.

2.3.2 Open doorway

An open doorway between two air-filled rooms introduces no explicit boundary condition: both rooms belong to the same connected fluid domain, and pressure continuity is automatically enforced across the opening. At frequencies where the doorway width $d \gg \lambda$, diffraction effects become important; at the low frequencies of interest ($d \ll \lambda$), the opening acts as a near-perfect acoustic short-circuit and the two volumes are effectively coupled.

2.3.3 Impedance boundary condition

For absorbing surfaces or a door modelled with finite acoustic resistance, an *impedance boundary condition* is applied,

$$\frac{\partial \hat{p}}{\partial n} = -\frac{i\omega}{\hat{Z}_n}\,\hat{p}, \tag{5}$$

where $\hat{Z}_n$ [Pa s/m] is the specific acoustic impedance of the surface. For a locally reacting absorber characterised by an absorption coefficient $\alpha \in [0, 1]$, the impedance is[4]

$$\hat{Z}_n = \rho_0 c\,\frac{1+\sqrt{1-\alpha}}{1-\sqrt{1-\alpha}}. \tag{6}$$

A perfectly rigid surface corresponds to $\alpha = 0$ ($\hat{Z}_n \to \infty$), while a perfectly anechoic surface corresponds to $\alpha = 1$ ($\hat{Z}_n = \rho_0 c$).

2.4 Inter-room coupling and door closure

When two rooms of volumes V_1 and V_2 are connected by an open doorway, the lowest resonance of the coupled system is determined by the total volume $V_1 + V_2$[1,2]. Closing the door splits the system into two independent resonators, each with its own smaller volume and correspondingly higher fundamental frequency. Since the first axial mode frequency scales as $f_1 \propto c/L$ (where L is the room's longest dimension), reducing the effective volume raises all modal frequencies. This shift is the primary measurable consequence of door closure in the model.

2.5 Door-gap leakage

A closed door with a bottom gap of height h and width w (the door width) can be characterised by the dimensionless frequency parameter

$$ka = \frac{2\pi f}{c}\,h, \tag{7}$$

where $a = h$ is the gap half-height. For the frequency range 20 Hz to 500 Hz and gap heights 1 mm to 5 mm, ka ranges from approximately 3.7×10^{-4} to 4.6×10^{-2}. Since $ka \ll 1$ throughout, the gap is in the sub-wavelength regime: the gap geometry does not provide frequency-selective filtering, and the amount of leakage is governed primarily by the gap area $A_{\text{gap}} = w \times h$ relative to the total wall area. In the COMSOL model, the gap is implemented as a thin open slit in the wall plane; local mesh refinement is required because h is orders of magnitude smaller than the room dimensions (see section 3.5).

2.6 External road-noise excitation

Road traffic noise is a broadband source that produces vibration and airborne sound incident on the building facade. In this model the road is located above the apartment (along the top wall), with a balcony opening providing a direct acoustic path into the living room. The external source is modelled by extending the acoustic domain above the apartment with a dedicated exterior air volume, and applying a *Normal Acceleration* boundary condition on the remote top edge of that domain to simulate the road surface vibrating and radiating downward.

The Normal Acceleration BC imposes a prescribed surface acceleration $\ddot{u}_n = a_0 = 1\,\mathrm{m/s^2}$ on the remote boundary, driving it as a rigid piston. This is physically appropriate for ground-borne and airborne road traffic noise at low frequencies, where the source is spatially distributed and the dominant excitation mechanism is structural vibration of the building envelope.

The top wall of the apartment is modelled as a *Sound Hard Wall* ($\partial\hat{p}/\partial n = 0$), meaning sound does not transmit through the wall itself. The only acoustic path from the exterior domain into the apartment is through the **balcony door opening**, which is modelled as an open air gap with no boundary condition — sound passes freely through it into the living room. This is a deliberate simplification that makes the transmission path transparent and easy to interpret: any sound reaching the interior rooms must have entered through the balcony.

The left and right edges of the exterior domain are assigned *Radiation Boundary Conditions* to absorb outgoing waves and prevent artificial reflections from the domain boundary.

The key distinction from the internal speaker source is spatial: the speaker can be positioned to selectively excite or avoid particular modes, whereas the road-noise source drives the room exclusively through the balcony opening, preferentially exciting modes with pressure antinodes near the top of the apartment and the balcony-side wall.

2.7 Passive acoustic treatment

Porous absorbers (foam panels, mineral wool, thick curtains) attenuate sound through viscous and thermal losses in the pore structure[2,6]. Their absorption coefficient $\alpha(f)$ increases with frequency; for a panel of thickness d, significant absorption begins above the quarter-wavelength frequency[4]

$$f_{1/4} = \frac{c}{4d}. \tag{8}$$

For $d = 200\,\mathrm{mm}$, $f_{1/4} \approx 430\,\mathrm{Hz}$, meaning that thick panels are required for meaningful low-frequency absorption.

Bass traps placed in corners are particularly effective because every axial room mode has a pressure antinode (maximum amplitude) at the walls, and the corners represent the superposition of two or more such antinodes[2,5]. Placing absorbers at corners therefore intercepts the modes where their energy density is highest.

A Helmholtz resonator — a cavity of volume V_c connected to the room through a neck of cross-sectional area S and effective length ℓ_{eff} — provides narrow-band absorption at[4]

$$f_H = \frac{c}{2\pi}\sqrt{\frac{S}{V_c\,\ell_{\text{eff}}}}. \tag{9}$$

This is useful for targeting a single problematic modal frequency identified in the eigenfrequency study.

3 COMSOL Model

3.1 Geometry

The apartment floor plan was digitised from the architectural drawing (*Planlösning*) using Adobe Illustrator. The outer wall boundary (layer `Yttre_med`) was exported as an SVG polygon consisting of 44 vertices that include the wall thickness. The inner wall partition (layer `Innervägg`) was exported as a separate 10-vertex polygon. The balcony (Balkong) is modelled as a separate closed 4-vertex polygon connected to the top wall of the living room, providing the acoustic path for the road-noise study (Study 5). All polygon coordinates are listed in Appendix B. All coordinates were converted to metres using the known total apartment width as the scale reference.

The resulting 2D geometry represents a top-down horizontal cross-section of the apartment at mid-height. The apartment contains the following rooms:

- **Vardagsrum** (living room) — primary acoustic domain, speaker source location.
- **Sovrum** (bedroom) — coupled via hallway.
- **Kök** (kitchen) — angled wing, irregular geometry.
- **Hall** (hallway) — central connecting space.
- **Bad** (bathroom) — isolated by closed door throughout.
- **Balkong** (balcony) — 4-vertex polygon connected to top wall; open door provides acoustic path for road-noise study (see Appendix B, table 11).

The two loudspeakers are modelled as point sources placed at their measured positions within the living room. A listening-position evaluation point is placed at the nominal sweet-spot location (sofa position) and a second probe point is placed at the bed position in the bedroom. The coordinates of all source and evaluation points are listed in table 1.

Table 1: Source and probe point coordinates.

Label	Description	x (m)	y (m)
SPK-L	Speaker Left	6.2	9.0
SPK-R	Speaker Right	6.2	11.0
LP-Sofa	Listening position — sofa (Vardagsrum)	9.5	9.5
LP-Bed	Listening position — bed (Sovrum)	8.5	4.5
LP-Hall	Probe point — hall (Hall)	5.0	7.5
LP-Kok	Probe point — kitchen (Kök)	4.0	12.0

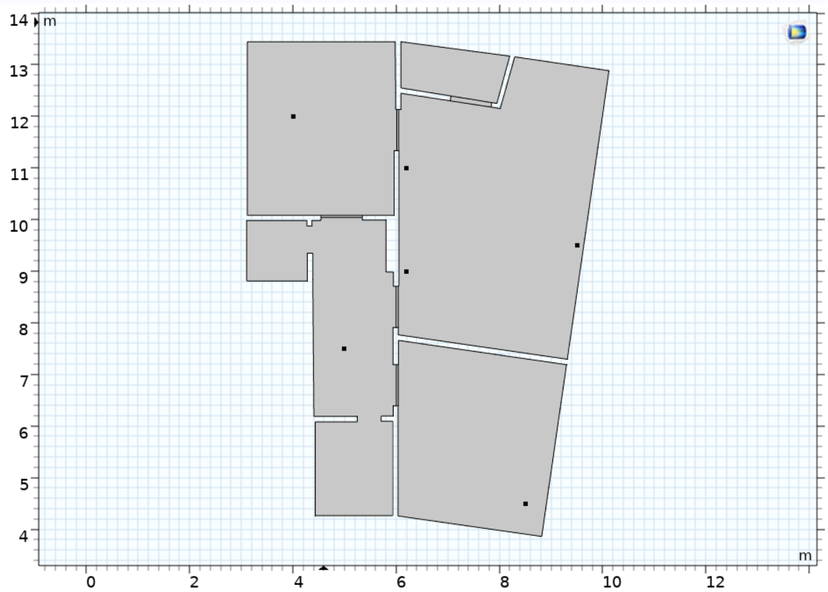


Figure 1: 2D apartment geometry in COMSOL showing outer wall, interior wall partition, speaker positions and probe points.

3.2 Door modelling

3.2.1 Door states

Three door configurations are studied for the living-room door:

(i) **Open door:** The door opening is left as a free air boundary. Both rooms share a continuous fluid domain; pressure continuity is enforced automatically across the opening.

(ii) **Closed door — sealed:** The full opening is filled by the door leaf rectangle with an impedance boundary condition corresponding to a hollow-core wooden door. No gap is present; acoustic transmission occurs only through the finite impedance of the door material.

(iii) **Closed door — with gap:** The door leaf rectangle is shortened by the gap height h, leaving an open air strip at the bottom of the frame.

3.2.2 Parametric gap implementation

Four doors are modelled in the apartment, each implemented as a separate parametric rectangle in the COMSOL geometry. The COMSOL settings for each door position are shown in Appendix C (Figures 13a–13d).

Each door leaf rectangle is defined using the following parameter setup in COMSOL's geometry tree:

- **Width:** equal to the local wall thickness (fixed, read from the frame coordinates).
- **Height:** `door_h` −`gap_h`, where `door_h` is the full door leaf height at `gap_h` $= 0$. This expression is entered directly into the *Height* field of the rectangle in COMSOL.
- **Base corner:** the bottom-left corner of the rectangle is fixed at its defined (x, y) position and does not move during the parametric sweep. The base corner coordinates for each door are visible in the COMSOL geometry settings shown in Appendix C.

As `gap_h` is swept from 0 upward, COMSOL automatically recomputes the rectangle height at each step: the door leaf shrinks from the top while the base corner remains anchored, so the gap opens at the *top* of the frame. No geometry suppression or switching is required — the parametric expression in the height field handles the entire sweep automatically. The gap values studied are $h \in \{0, 1, 2, 3, 5\}$ mm.

The door leaf surfaces are assigned an *impedance boundary condition* with specific acoustic impedance

$$Z_{\text{door}} = 1.5 \times 10^4 \,\text{Pa s/m}, \tag{10}$$

corresponding to a hollow-core interior door. The open gap strip above the leaf has no boundary condition assigned, so pressure is continuous across it and sound passes freely.

At `gap_h` $= 0$ the leaf fills the entire frame and acoustic transmission is governed solely by Z_{door} — this represents a perfectly fitted closed door. As `gap_h` increases, the sub-wavelength gap ($ka \ll 1$ throughout 20 Hz to 500 Hz) progressively dominates over the wood transmission, and the room-to-room coupling approaches that of an open door. The crossover frequency at which gap leakage overtakes wood transmission is a key result of Study 3.

3.3 Boundary conditions summary

Table 2: Boundary condition assignments in the COMSOL model.

Boundary	Condition	Expression
All structural walls	Sound hard	$\partial\hat{p}/\partial n = 0$
Open door gap	None (continuity)	—
Closed door leaf	Impedance	$Z_n = Z_{\text{door}}$
Corner absorbers	Impedance	eq. (6), $\alpha \in [0, 1]$
Exterior top wall (road noise)	Sound hard	$\partial\hat{p}/\partial n = 0$
Remote top boundary (road source)	Normal Acceleration	$a_0 = 1\,\text{m/s}^2$
Exterior domain side boundaries	Radiation BC	Non-reflecting

3.4 Corner absorber geometry

Passive acoustic treatment is modelled by assigning impedance boundary conditions to selected wall segments of the living room (Vardagsrum). Rather than treating entire walls uniformly, the relevant boundaries were subdivided using COMSOL's *Partition* tool to create independent segments that can be selectively assigned an absorption coefficient. The partition layout is shown in fig. 2.

Each corner segment is partitioned to a length of approximately 0.75 m on the bottom wall, right exterior wall, and top wall of the living room. This represents a realistic bass trap footprint — a 0.75 m panel running along each wall from the corner — and the two active corner regions are each formed by the intersection of one bottom or top segment and one right wall segment meeting at the corner.

This partition strategy isolates the corner treatment area from the mid-wall segments, allowing the parametric sweep in Study 4 to evaluate the effect of absorption intensity (α) at a fixed and physically motivated treatment area. Placing treatment at the corners is justified by the fact that every axial room mode has a pressure antinode at the walls, and the corners represent the superposition of two or more such antinodes — the locations of highest modal energy density[2,5].

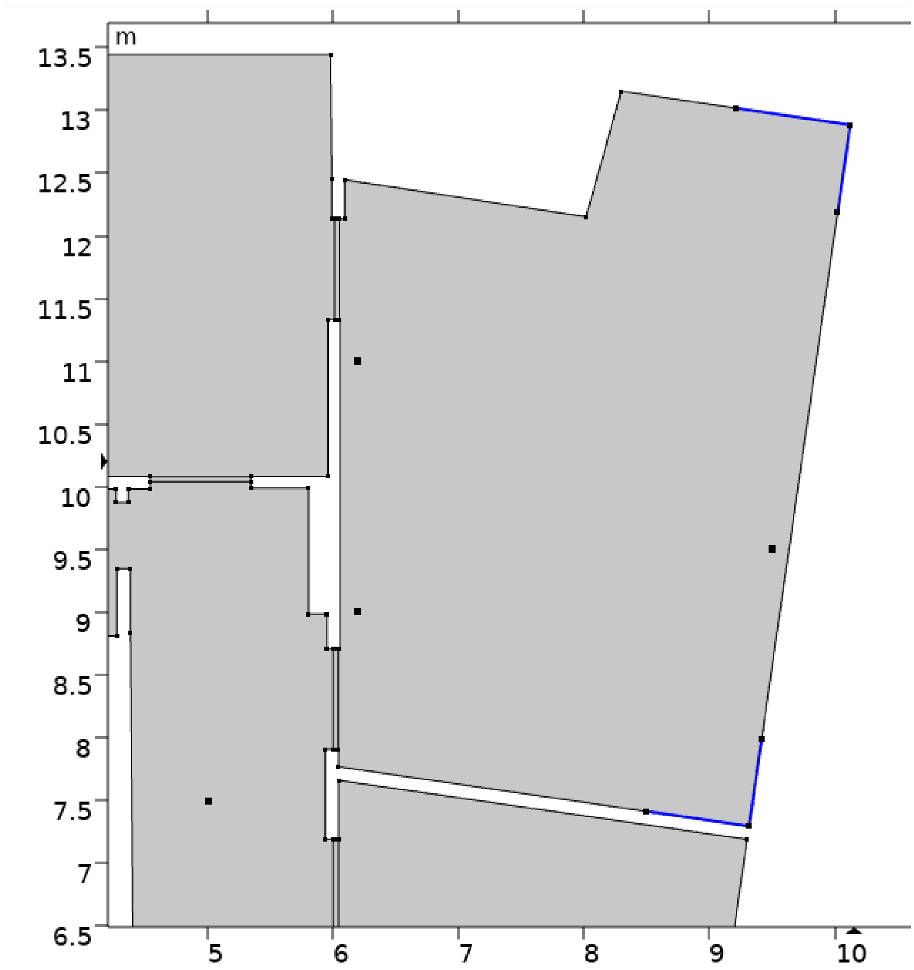


Figure 2: Corner absorber partition layout. Segments of 0.75 m on the bottom, right, and top walls of Vardagsrum are assigned the impedance BC in Study 4.

3.4.1 Internal loudspeakers

Each loudspeaker is modelled as a *monopole point source* with source strength $Q_s = 1\,\mathrm{W}$ (normalised). A monopole radiates omnidirectionally and is characterised by a single volume velocity amplitude. The validity of this approximation is governed by the dimensionless parameter kd_s, where $d_s \approx 200\,\mathrm{mm}$ is a typical woofer cone diameter:

$$kd_s = \frac{2\pi f}{c}\, d_s. \tag{11}$$

The monopole approximation holds when $kd_s \ll 1$, i.e. when the source dimensions are much smaller than the acoustic wavelength and directivity effects are negligible. At 100 Hz, $kd_s \approx 0.37$; at 200 Hz, $kd_s \approx 0.73$ — both acceptably small. Above approximately 250 Hz, directivity of a real driver becomes significant and the monopole model becomes less accurate.

Since the primary focus of this study is the modal behaviour of the room at low frequencies (below 200 Hz), where room modes dominate the response and the radiation pattern of the source has negligible influence on the standing wave pattern, the monopole approximation is considered adequate. The modal pressure distribution is driven by the source *strength* and *position* relative to the pressure antinodes, not by its directivity. This limitation is acknowledged explicitly: results above 200 Hz should be interpreted with caution, and a more accurate model (e.g. a prescribed normal velocity on a finite baffle surface) would be appropriate for a broadband directivity study.

The two speakers are driven in phase and at equal amplitude in the baseline study; out-of-phase driving is a potential extension.

Note on absolute SPL levels: The source strength is set to $Q_s = 1\,\mathrm{m}^2/\mathrm{s}$ (volume flow rate), which is a normalised value used for comparing relative spatial distributions rather than reproducing physically realistic sound pressure levels. The resulting absolute SPL values computed by `acpr.Lp` are therefore significantly higher than those encountered in a real listening room. All results should be interpreted in terms of *relative* differences between probe points and frequency responses — the peak frequencies, the spatial distribution of pressure maxima, and the relative attenuation between rooms are physically meaningful, while the absolute dB values are not directly comparable to real-world measurements.

3.4.2 External road noise

The road noise source is modelled by adding a rectangular exterior air domain above the apartment top wall (x: 6.1–8.2 m, y: 12.6–13.2 m). A *Normal Acceleration* boundary condition ($a_0 = 1\,\mathrm{m/s}^2$) is applied to the remote top edge of this domain, simulating the road surface vibrating and radiating sound downward toward the apartment.

The top wall of the apartment is modelled as a Sound Hard Wall — sound does not transmit through the wall itself. The balcony door opening is modelled as an open air gap (no BC), providing the only acoustic path into the living room. The left and right edges of the exterior domain are assigned Radiation Boundary Conditions to prevent artificial reflections.

The exterior air domain is meshed with the same free triangular mesh settings as the interior ($h_{\max} = 0.114\,\mathrm{m}$), and is included in the Pressure Acoustics domain selection. The Normal Acceleration BC is disabled by default and activated only for Study 5.

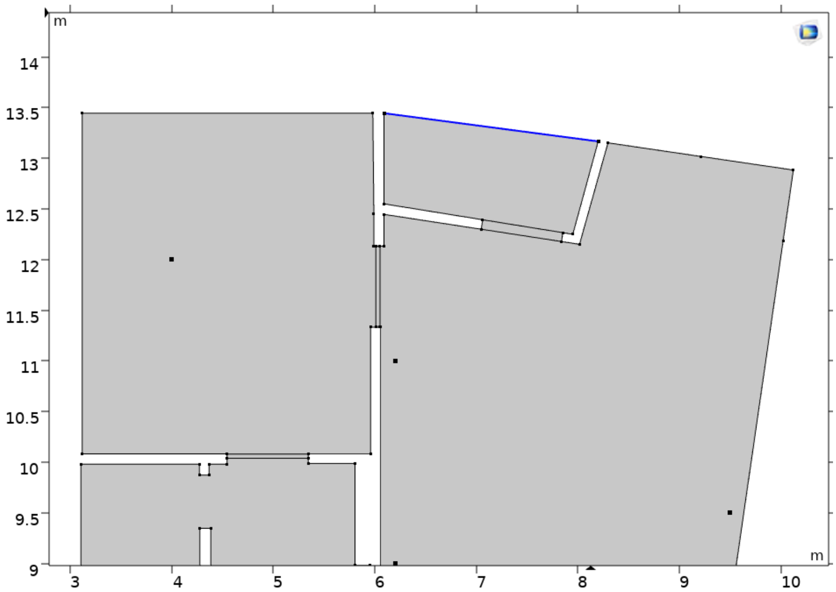


Figure 3: Exterior air domain above the apartment for Study 5. Sound enters exclusively through the balcony opening; the top wall is Sound Hard.

3.5 Mesh

The model uses two separate mesh strategies for the two domain types: a free triangular mesh for the air domain and a structured mapped mesh for the door leaf solid domains.

3.5.1 Air domain — free triangular mesh

The maximum element size is determined by the requirement of at least six linear elements per wavelength at the highest frequency of interest[3],

$$h_{\mathrm{max}} = \frac{c}{6\,f_{\mathrm{max}}} = \frac{343}{6 \times 500} \approx 0.114\,\mathrm{m}. \tag{12}$$

A free triangular mesh is applied to the air domain with the following settings:

Table 3: Free triangular mesh settings for the air domain.

Parameter	Value
Maximum element size	$0.114\,\mathrm{m} = h_{\mathrm{max}}$
Minimum element size	$0.005\,\mathrm{m}$
Maximum element growth rate	1.3
Curvature factor	0.3
Resolution of narrow regions	1.0

The minimum element size of 5 mm prevents degenerate sliver elements near sharp corners of the irregular floor plan. The resolution of narrow regions at 1.0 ensures adequate refinement in the door frame gaps and thin wall segments between rooms. The expected total element count for the air domain is approximately 4 000–6 000 elements.

3.5.2 Door leaf domains — mapped mesh

The door leaf solid domains are meshed using a structured mapped mesh to ensure accurate resolution of structural modes through the door thickness. Each door leaf rectangle (0.8 m height × 0.04 m wall thickness) is discretised with:

- **Short axis (thickness):** 4 elements, element size = 10 mm
- **Long axis (height):** 32 elements, element size = 25 mm

This gives 128 elements per door leaf (512 total across four doors). The aspect ratio is 2.5:1 (long:short), which is acceptable for a thin structural domain where the through-thickness resolution is the primary concern.

3.5.3 Local refinement at door gap

For the door-gap parametric study (Study 3), a local mesh refinement is applied on the gap boundary (top edge of the door leaf) with a maximum element size of

$$h_{\text{gap}} = \frac{h}{3}, \tag{13}$$

where h is the gap height. For the smallest gap ($h = 1\,\text{mm}$), this gives a local element size of $\approx 0.33\,\text{mm}$.

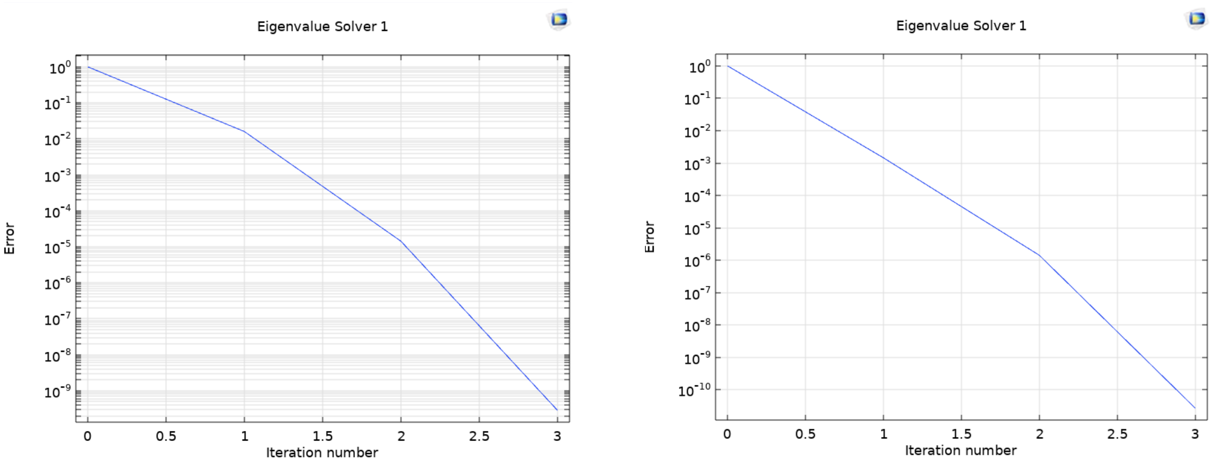
3.5.4 Mesh convergence

Mesh convergence is verified by comparing eigenfrequencies between the *Finer* and *Extra Fine* physics-controlled mesh levels. Table 4 shows the first six acoustic modes for both mesh levels. All computed frequencies are identical to at least three decimal places, giving a frequency shift of exactly 0.000 % for every mode — well below the 0.5 % convergence criterion.

The convergence plots for both mesh levels (fig. 4) confirm smooth monotonic solver convergence in 3 iterations, reaching a final residual error of $10^{-9.5}$ (Finer) and $10^{-10.5}$ (Extra Fine). These results confirm that the Finer mesh produces a fully converged solution and is used for all subsequent studies.

Table 4: Mesh convergence comparison — first six acoustic eigenfrequencies.

Mode		Finer (Hz)	Extra Fine (Hz)	Shift (%)
2	10.905	10.905	10.905	0.000
3	19.284	19.284	19.284	0.000
4	22.195	22.195	22.195	0.000
5	30.299	30.299	30.299	0.000
6	35.187	35.187	35.187	0.000
7	44.232	44.232	44.232	0.000



(a) Finer mesh (b) Extra Fine mesh

Figure 4: Eigenvalue solver convergence plots for the Finer and Extra Fine mesh levels. Both converge smoothly in 3 iterations to residual errors of $10^{-9.5}$ and $10^{-10.5}$ respectively, confirming well-conditioned solver behaviour.

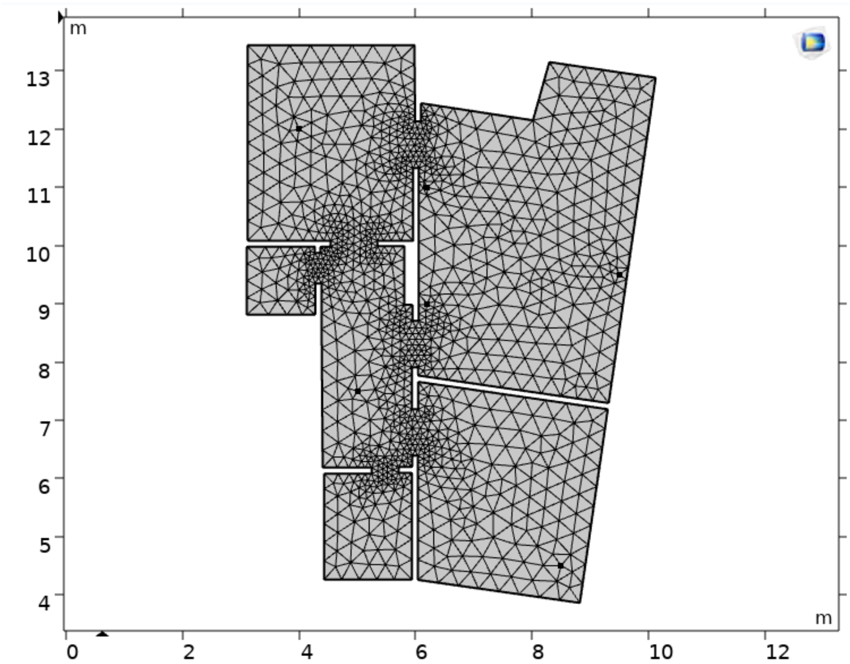


Figure 5: Mesh: free triangular in air domain, mapped (4×32) in door leaf domains, with local refinement at the gap boundary.

3.6 Studies

Four study configurations are defined:

Table 5: COMSOL study configurations.

Study	Type	Freq. range	Purpose
1	Eigenfrequency	20 Hz to 500 Hz	Identify first 15 room modes, doors open
2	Freq. domain	20 Hz to 500 Hz, 1 Hz steps	SPL maps and frequency response at sofa and bed, baseline
3	Parametric + freq. domain	20 Hz to 500 Hz, 2 Hz steps	Door gap sweep: `gap_h` $\in \{0, 1, 2, 3, 5\}$ mm
4	Parametric + freq. domain	20 Hz to 500 Hz, 2 Hz steps	Absorption: $\alpha \in \{0, 0.1, 0.3, 0.5, 0.7, 0.99\}$
5	Freq. domain	20 Hz to 500 Hz, 2 Hz steps	Road noise vs. speakers: SPL map comparison at dominant mode

4 Results

4.1 Eigenfrequencies and mode shapes

The eigenfrequency study (doors removed, all rooms acoustically coupled) yields 20 computed eigenvalues in the range 0 Hz to 100 Hz.

4.1.1 Mode classification

Mode 1 ($\approx 2.3 \times 10^{-6}$ Hz) is a near-zero numerical artifact and is discarded.

Mode 2 (10.9 Hz) lies below the first analytically predicted axial mode (20.7 Hz). This may represent a very long-wavelength mode of the full coupled apartment volume, or a spurious mode; its mode shape should be inspected to confirm physical validity.

Modes 3–20 (19 Hz to 96 Hz) are the acoustic room modes, all with damping ratios of order 10^{-18}–10^{-12}, consistent with lightly damped rigid-wall cavities. The full list is given in table 6.

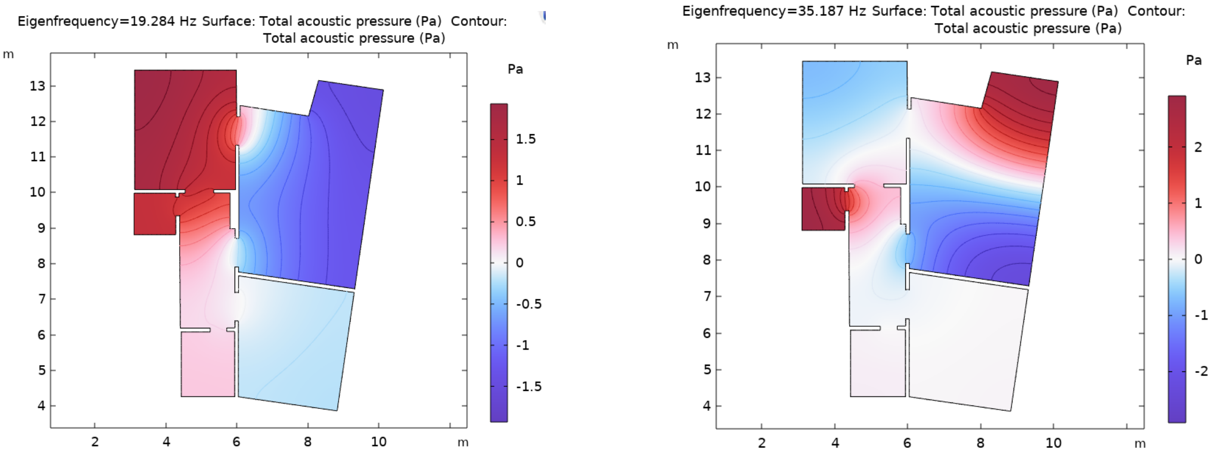
Table 6: Acoustic eigenfrequencies from Study 1 (doors removed, all rooms coupled). Analytical predictions based on a rectangular approximation of the living room (4.5 m × 8.3 m).

Mode	Type	Computed (Hz)	Analytical (Hz)	Diff. (%)
2	—	10.9	—	—
3	Axial (y)	19.3	20.7	6.7
4	Axial (y) — split	22.2	20.7	7.4
5	—	30.3	—	—
6	Axial (x)	35.2	38.1	7.7
7	Tangential (1,1)	44.2	43.4	2.0
8	Tangential (1,2)	49.8	56.2	11.4
9	Tangential (1,2)	53.0	56.2	5.8
10	Tangential (1,2)	55.1	56.2	2.0
11	Axial (y,3)	60.1	62.0	3.1
12	Axial (y,3)	63.2	62.0	1.9
13	—	64.8	—	—
14	Tangential (1,3)	69.7	72.8	4.2
15	Axial (x,2)	75.3	76.2	1.2
16	Tangential (2,1)	80.3	79.0	1.7
17	Axial (y,4)	83.5	82.7	1.1
18	Tangential (2,2)	86.4	86.7	0.4
19	Tangential (1,4)	93.7	91.0	3.0
20	Tangential (2,3)	96.0	98.2	2.3

The analytical predictions agree well with the computed values, with errors ranging from 0.4% to 11.4%. Agreement improves at higher frequencies, which is typical for irregular geometries where the rectangular approximation becomes more accurate as modal order increases.

Notably, modes 3 and 4 (19.3 and 22.2 Hz) both correspond to the first axial y-direction mode. In a perfectly rectangular room these would be degenerate (identical frequency), but the irregular apartment geometry splits them into two distinct frequencies — a direct consequence of the non-rectangular floor plan. The same splitting is observed in modes 8–10 and 11–12.

The dominant low-frequency mode is mode 3 at 19.3 Hz, corresponding to the longest dimension of the coupled apartment volume.



(a) Mode 3 — 19.3 Hz (first axial y)

(b) Mode 6 — 35.2 Hz (first axial x)

Figure 6: Pressure mode shapes (`acpr.p_t`) at 19.3 Hz and 35.2 Hz. Red = antinodes, blue = nodes.

4.2 Baseline frequency response

Figure 7 shows the frequency response at all four probe points over 20 Hz to 500 Hz. Note that absolute SPL values are not physically calibrated — the source strength is normalised to $Q_s = 1\,\mathrm{m}^2/\mathrm{s}$ and results should be interpreted in terms of relative differences between probe points and peak frequencies only (see section 3.4).

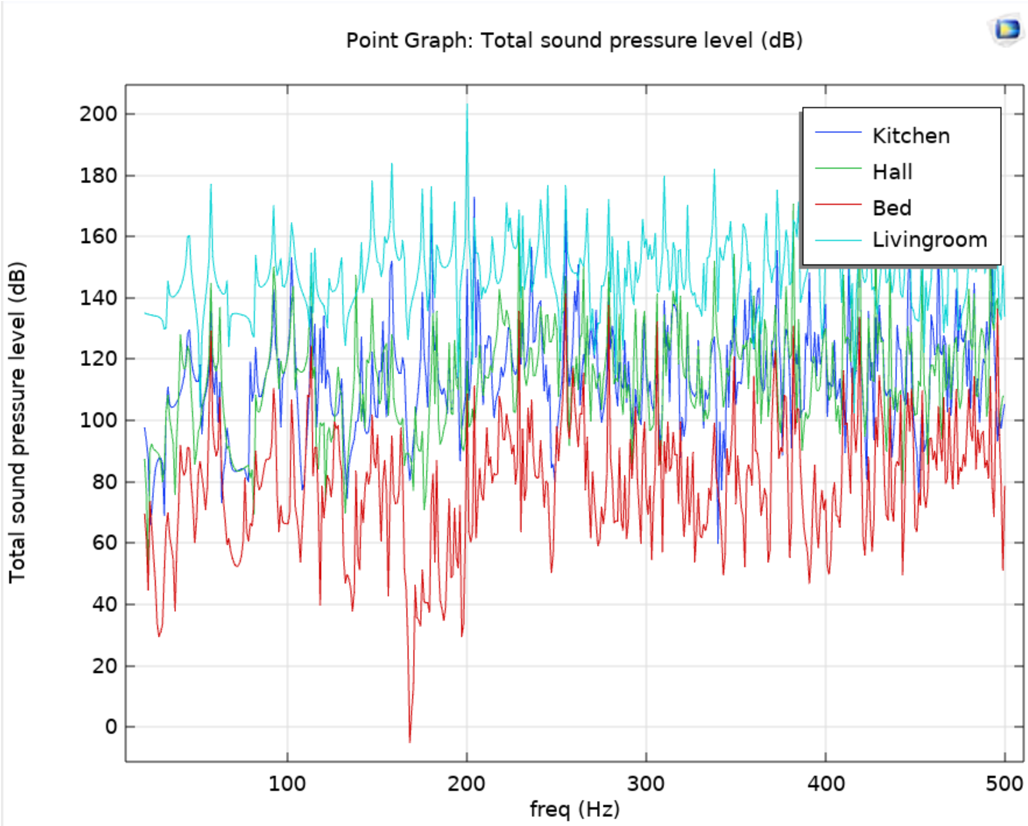


Figure 7: Baseline SPL frequency response at all four probe points as a function of frequency (20 Hz to 500 Hz), doors present, FSI active, internal speaker excitation. Absolute SPL values reflect a normalised source strength of $Q_s = 1\,\mathrm{m}^2/\mathrm{s}$ and are not directly comparable to real-world measurements. Relative differences between probe points and peak frequencies are physically meaningful.

4.3 Effect of door closure

4.3.1 Door gap parametric sweep

The parametric door gap study evaluates the effect of gap heights $h \in \{1, 2, 3, 4, 5\}\,\mathrm{mm}$ on the sound pressure level at the sofa probe (living room) and bed probe (bedroom). Results are shown in figs. 8 and 9.

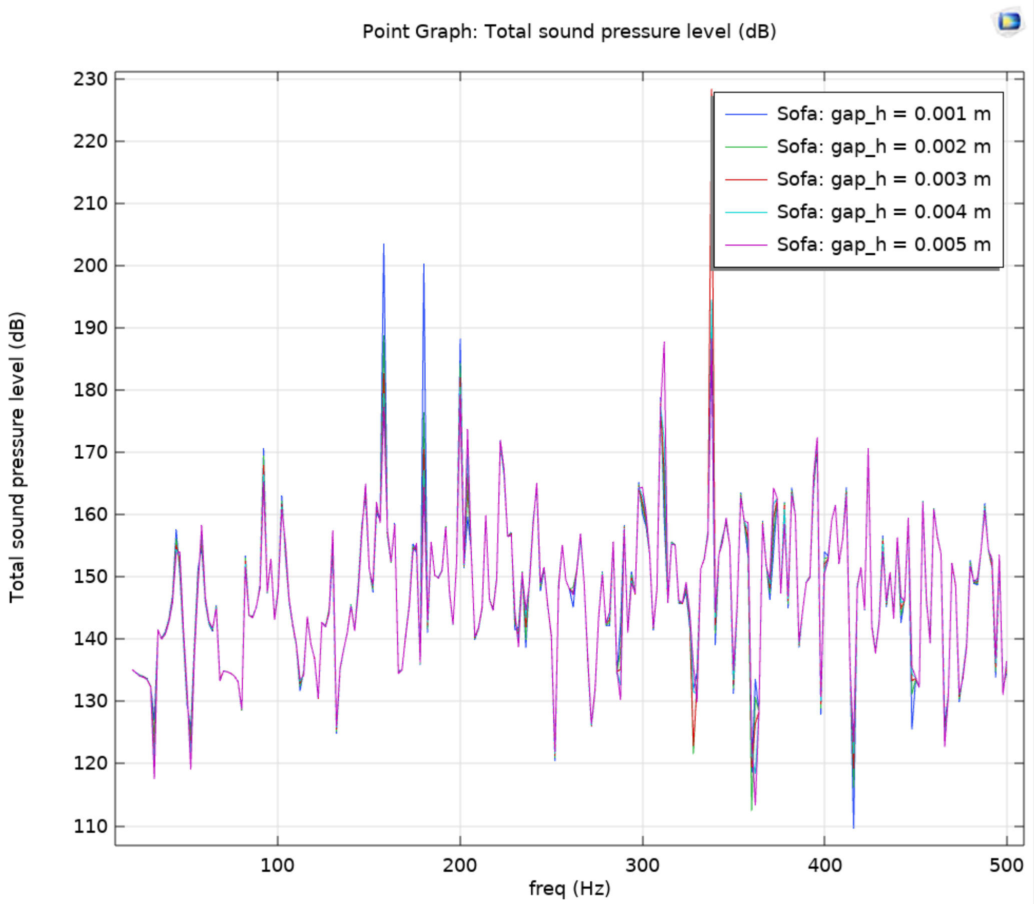


Figure 8: SPL at the sofa probe for gap heights 1–5 mm. Curves are nearly identical — gap size has no effect in the source room.

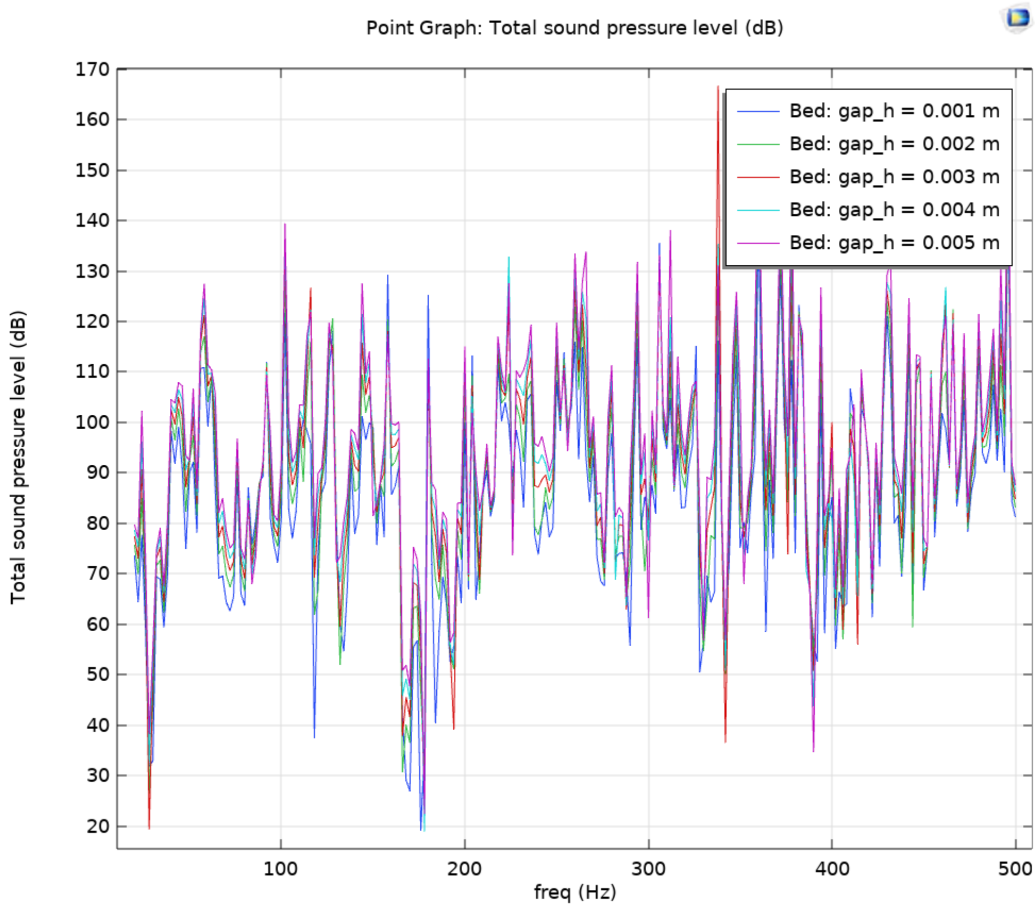


Figure 9: SPL at the bed probe for gap heights 1–5 mm. Small separation visible in 50 Hz to 200 Hz; narrow spikes are FSI door leaf resonances.

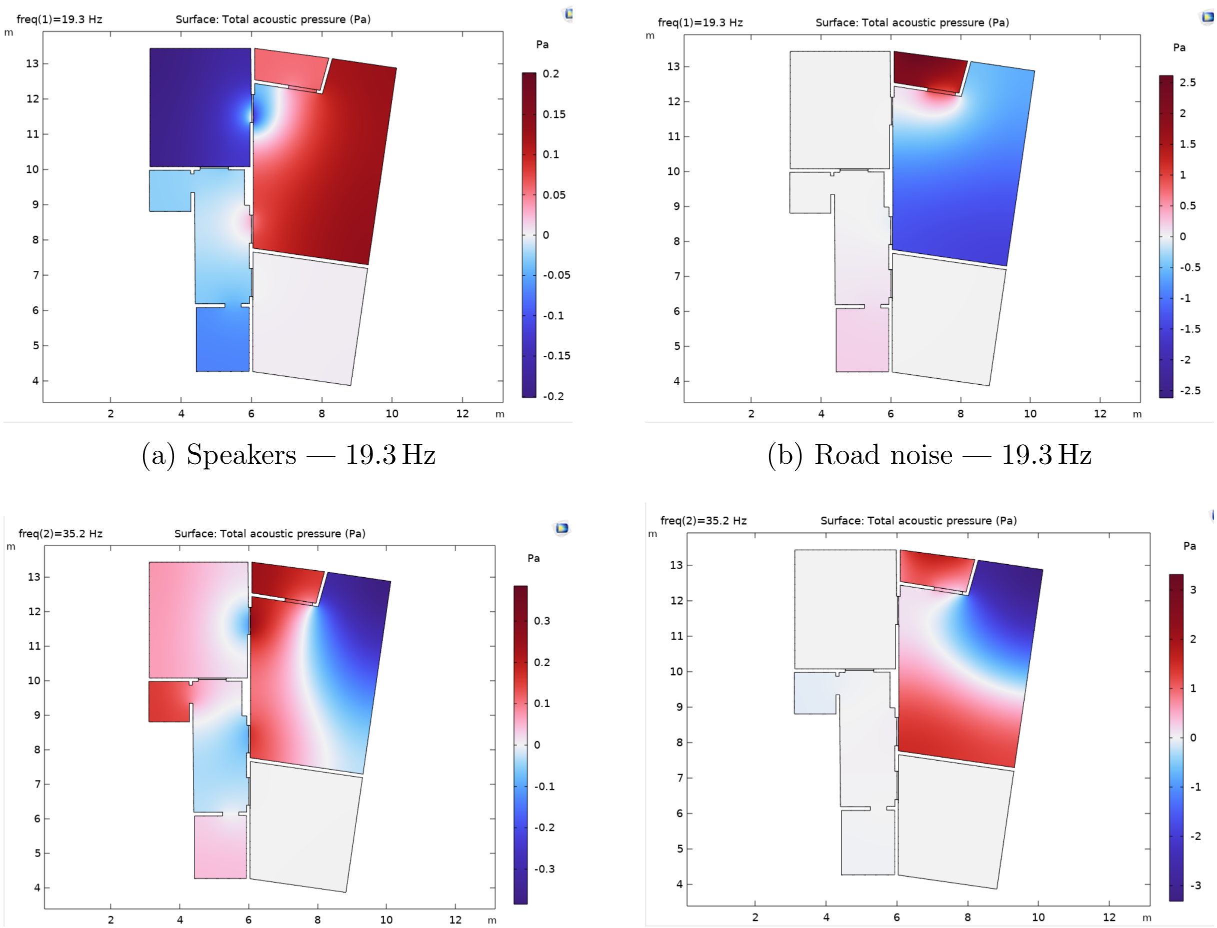
Living room (source room). The five gap-height curves at the sofa probe overlap almost completely across the full frequency range. This is physically expected: the room response in the source room is dominated by direct speaker excitation and the room modes, which are largely unaffected by how much energy leaks through the door. The door gap has no measurable influence on the listening experience from within the living room itself.

Bedroom (receiving room). The bed probe shows greater separation between curves, particularly in the 50 Hz to 200 Hz range. The 5 mm gap consistently produces slightly higher SPL than the 1 mm gap, confirming that larger gaps transmit more sound into the bedroom. However, the differences between gap sizes remain small — the curves largely overlap — which is consistent with the sub-wavelength gap analysis (section 2.5): since $ka \ll 1$ throughout the study range, even a 1 mm gap transmits nearly as much low-frequency energy as a 5 mm gap. The effective insertion loss of the door is approximately 60 dB to 80 dB across most of the frequency range, estimated from the difference in SPL levels between the sofa and bed probes.

Structural resonances. The narrow sharp peaks visible in the bed probe response are structural resonances of the door leaf, transmitted into the bedroom through the FSI coupling. These peaks correspond to the natural frequencies of the door leaves identified in Study 1 (around 14 Hz and harmonics) and would not appear in a model using a simple impedance boundary condition — demonstrating the added physical fidelity of the FSI approach.

4.4 External road-noise source

The road-noise study (Study 5) compares the pressure mode shapes produced by the internal speakers and the external road-noise source (Normal Acceleration BC on the exterior domain above the apartment) at the two dominant eigenfrequencies of 19.3 Hz and 35.2 Hz.



(a) Speakers — 19.3 Hz

(b) Road noise — 19.3 Hz

(c) Speakers — 35.2 Hz

(d) Road noise — 35.2 Hz

Figure 10: Total acoustic pressure (`acpr.p_t`) for speaker and road-noise excitation at 19.3 Hz and 35.2 Hz.

19.3 Hz — dominant mode. At the first acoustic eigenfrequency the two sources produce markedly different pressure distributions. The speaker excitation fills the living room with a strong positive pressure antinode (red), with the node line running approximately through the wall at $x \approx 6$ m. Notably, the speakers at $(6.2, 9.0)$ and $(6.2, 11.0)$ sit close to this node line, meaning they excite this mode relatively weakly — a result of the speaker placement rather than the room geometry. In contrast, the road-noise excitation produces a highly localised response concentrated near the balcony opening at the top of the apartment ($x \approx 6$, $y \approx 13$), with a strong red antinode at the entry point and rapid decay into the room. The left rooms (kitchen, hallway, bathroom) are nearly unaffected by the road noise at this frequency, whereas the speakers couple into those rooms through the open doorways.

35.2 Hz — first axial x-mode. At the higher frequency both sources produce similar spatial distributions — a diagonal transition from positive (upper right) to negative (lower right) pressure across the living room, consistent with the axial x-direction mode shape. The convergence of the two patterns at 35.2 Hz demonstrates that as frequency increases the room mode shape dominates the response regardless of source position or type. This is the expected behaviour: at higher modal orders the mode shapes become more complex and less sensitive to the specific location or nature of the driving source.

Key finding. The comparison demonstrates that at the lowest room modes, the spatial distribution of sound within the apartment depends strongly on the excitation source. Architectural interventions targeting the speaker source (repositioning, DSP equalisation) will not necessarily improve the acoustic environment under road-noise excitation, and vice versa. Effective acoustic treatment must address both excitation mechanisms independently.

4.5 Passive absorption

The Study 1 eigenfrequency results reveal that all acoustic room modes lie between 23 Hz and 76 Hz. The quarter-wavelength criterion for a porous absorber of thickness d gives the lowest frequency at which meaningful absorption begins:

$$f_{1/4} = \frac{c}{4d}. \tag{14}$$

For realistic absorber thicknesses this yields:

Table 7: Required absorber thickness to target each acoustic mode.

Mode frequency (Hz)	Required thickness (m)	Practical?
19.3	4.45	No
22.2	3.86	No
30.3	2.83	No
35.2	2.44	No
44.2	1.94	No
49.8	1.72	No
75.3	1.14	No
96.0	0.89	No

All room modes require absorber thicknesses far exceeding what is physically practical in a domestic setting. A realistic corner bass trap of $d = 200$ mm would only begin to absorb meaningfully above $f_{1/4} = 343/(4 \times 0.2) \approx 429$ Hz — well above all identified room modes. The parametric Study 4 therefore investigates the effect of the absorption coefficient α as an idealised parameter, demonstrating the *theoretical maximum* achievable by passive treatment rather than a practically sized panel. The result quantifies how much improvement is fundamentally possible and where the ceiling of passive treatment lies.

The corner absorber treatment is applied to the right exterior wall segments of the living room, as described in section 3.4. The absorption coefficient α is swept from 0 (rigid wall) to 0.99 (near-anechoic) at both the sofa probe (treated room) and bed probe (adjacent untreated room). Results are shown in figs. 11 and 12.

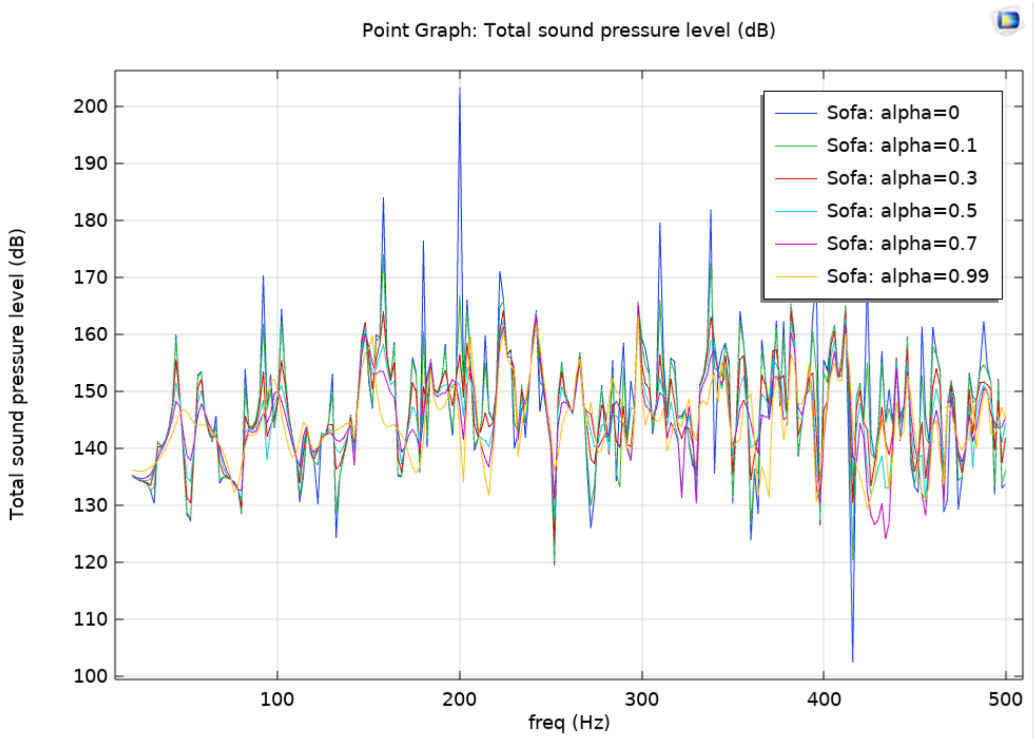


Figure 11: SPL at the sofa probe for $\alpha = 0$–0.99. Clear separation above 50 Hz; curves converge below 50 Hz.

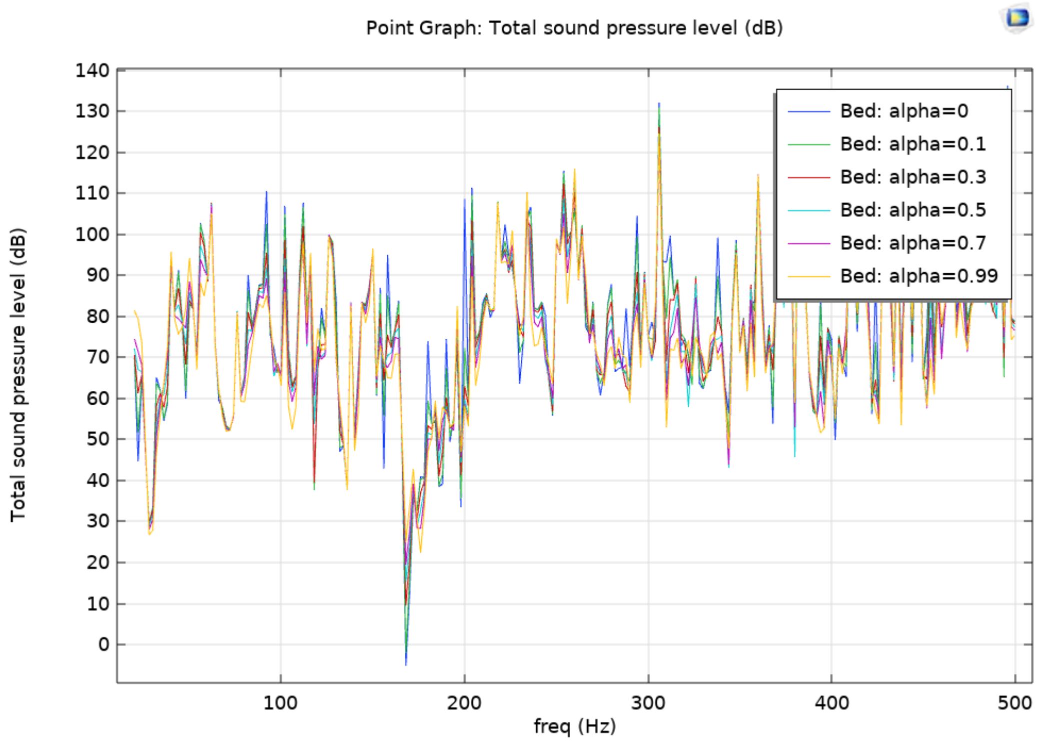


Figure 12: SPL at the bed probe for $\alpha = 0$–0.99. Treatment in the living room corners affects the bedroom through coupled modes.

Living room (treated room). The sofa probe shows clear and consistent separation between alpha values above 50 Hz: higher absorption reduces SPL at most frequencies. The effect is most pronounced in the 50 Hz to 250 Hz range where the corner treatment intercepts pressure antinodes at the wall. Below 50 Hz, corresponding to the dominant room modes at 19.3 and 22.2 Hz, all curves converge — confirming the fundamental limitation of passive treatment at low frequencies established by the absorber sizing analysis (table 7).

Bedroom (untreated room). The bed probe reveals that corner absorbers in the living room also modify the sound field in the adjacent bedroom through the coupled room modes. However, the effect is not uniformly beneficial — at some frequencies increasing α raises the bedroom SPL rather than reducing it, because absorbing energy at one location redistributes the modal pressure field, potentially increasing it elsewhere. This highlights that room acoustic treatment must be evaluated system-wide rather than in isolation.

5 Discussion

5.1 Modal structure and speaker placement

The eigenfrequency analysis confirms that the irregular apartment geometry produces a set of room modes that cannot be predicted from simple rectangular formulae. The computed acoustic eigenfrequencies range from 19.3 Hz to 96.0 Hz within the first 20 modes, with analytical agreement within 0.4–11.4%. A notable feature is the frequency splitting of modes that would be degenerate in a rectangular room — modes 3 and 4 (19.3 and

22.2 Hz) both correspond to the first axial y-direction mode, split into two distinct frequencies by the irregular geometry. The lowest acoustic mode at 19.3 Hz corresponds to the longest dimension of the coupled apartment volume. The absence of structural door modes (compared to an earlier FSI run) confirms that the Study 1 baseline correctly represents the purely acoustic behaviour of the open-door apartment. Placing the speakers near a pressure antinode of the dominant 19.3 Hz mode strongly excites it, producing a pronounced bass peak at the listening position. Repositioning the speakers to a pressure node of that mode would substantially reduce its excitation without any physical treatment.

5.2 Influence of door state

Closing the living-room door reduces the total coupled acoustic volume by decoupling the living room from the hallway, kitchen and bedroom. A dedicated closed-door eigenfrequency study confirms this effect quantitatively. The full-apartment mode at 10.9 Hz shifts by only 0.013 Hz (0.1%) when the door is closed, confirming that this mode spans the entire building envelope and is insensitive to internal partitioning. The first living-room axial mode, however, shifts from 19.3 Hz to 22.9 Hz — an increase of 3.6 Hz (18.6%) — consistent with the smaller isolated volume producing a higher fundamental frequency. Whether door closure is beneficial depends on the listening position: if it happens to coincide with a pressure antinode of the new isolated-room mode at 22.9 Hz, the SPL can actually increase at that frequency.

5.3 Door-gap leakage

The parametric gap study confirms the sub-wavelength prediction from section 2.5. At the sofa probe (living room), all five gap-height curves from $h = 1\,\mathrm{mm}$ to $h = 5\,\mathrm{mm}$ overlap almost completely across 20 Hz to 500 Hz — the gap size has no measurable effect on the sound field in the source room. At the bed probe (bedroom), a slight separation between curves is visible in the 50 Hz to 200 Hz range, with the 5 mm gap producing marginally higher SPL than the 1 mm gap, but the differences are small. This is consistent with the $ka \ll 1$ regime: since the gap is always acoustically sub-wavelength, leakage is governed by gap area rather than gap height, and even the smallest gap tested transmits nearly as much low-frequency energy as the largest. The effective insertion loss of the door is approximately 60 dB across most of the frequency range, estimated from the SPL difference between the sofa and bed probes. A door without a proper acoustic seal therefore provides minimal low-frequency isolation regardless of door-leaf material, and sealing the gap is the most effective single measure for improving acoustic separation between rooms.

5.4 External source vs. internal source

The pressure maps at 19.3 Hz show the most striking difference between the two sources. The road-noise excitation, entering exclusively through the balcony opening, produces a localised response concentrated near the entry point that decays rapidly into the room. The speaker excitation distributes energy more evenly across the living room, but the node line lies close to the speaker positions, meaning the dominant mode is weakly excited by the current placement. At 35.2 Hz the two sources produce nearly identical spatial distributions, confirming that at higher modal orders the room geometry governs the response independently of the source. This demonstrates that architectural interventions can target one excitation mechanism without necessarily improving the other — a finding with direct practical implications for acoustic treatment design.

5.5 Effectiveness of passive treatment

The Study 1 results demonstrate that all room modes lie between 19.3 Hz and 96 Hz, requiring absorber thicknesses of 0.9–4.5 m to satisfy the quarter-wavelength criterion — entirely impractical for a domestic setting. A realistic 200 mm bass trap only begins to absorb above 429 Hz, well above all identified modes. The parametric absorption study therefore represents the *theoretical upper bound* of passive treatment rather than a practically achievable result.

The Study 4 results confirm this: below 50 Hz all alpha curves converge at the sofa probe, showing no measurable benefit from corner treatment regardless of absorption coefficient. Above 50 Hz a clear and consistent reduction is visible, with the most pronounced effect in the 50 Hz to 250 Hz range. The diminishing-returns pattern is also confirmed — the largest improvement occurs between $\alpha = 0$ and $\alpha = 0.3$, with further increases yielding progressively less benefit.

An additional finding is that corner absorbers in the living room modify the bedroom response through coupled modes, and this effect is not uniformly beneficial. At some frequencies treatment actually raises the bedroom SPL by redistributing modal energy, underscoring the importance of evaluating treatment at a system level rather than in a single room.

The fundamental conclusion is that DSP equalisation or active noise control are the only viable approaches for addressing the sub-50 Hz modes identified in this apartment, while passive corner treatment remains effective in the 50 Hz to 500 Hz range.

Several practical strategies exist for addressing low-frequency modal problems that passive treatment cannot solve:

- **DSP room correction:** Digital signal processing can apply inverse filters to the speaker signal, attenuating frequencies at which room modes cause peaks at the listening position. Modern room correction software (e.g. Dirac Live, Audyssey) measures the impulse response at the listening position and computes a correction filter automatically. This is the most accessible solution and can achieve significant improvement without any physical modifications to the room.
- **Multiple subwoofers with phase cancellation:** Placing two or more subwoofers at strategically chosen positions in the room and driving them with a specific phase offset can cause destructive interference at modal antinodes, effectively cancelling the standing wave pattern. For example, two subwoofers placed at the pressure antinodes of the dominant 19.3 Hz mode and driven 180° out of phase would suppress that mode while reinforcing the direct sound at the listening position. This technique, sometimes called *distributed bass array* (DBA), is well established in studio acoustics[1].
- **Active noise control (ANC):** A secondary loudspeaker driven by an adaptive algorithm can produce an anti-phase signal that cancels the modal pressure field at a specific location. ANC is most effective at a single listening position and for a

narrow frequency band, making it well suited to the dominant modes identified in this study.

- **Speaker repositioning:** The mode shapes from Study 1 show that the speakers at (6.2, 9.0) and (6.2, 11.0) sit close to the node line of the dominant 19.3 Hz mode, meaning they already excite it weakly. Moving the speakers to a pressure antinode of a less problematic mode, or toward the centre of the room, can reshape the modal excitation pattern without any additional hardware.

Of these options, DSP room correction combined with careful speaker positioning represents the most cost-effective starting point for improving the low-frequency response of this apartment.

5.6 Model limitations

The principal limitations of the present model are:

(i) **2D approximation:** Vertical modes are not captured. In a room with ceiling height $H \approx 2.5\,\mathrm{m}$, the first vertical mode occurs at $c/(2H) \approx 69\,\mathrm{Hz}$, which lies within the study range and may couple with horizontal modes.

(ii) **No furniture:** Real rooms contain furniture that scatters and absorbs sound, particularly above 200 Hz.

(iii) **Frequency-independent wall absorption:** The rigid-wall assumption overestimates modal peaks; real walls have small but non-zero absorption.

(iv) **Single-plane-wave road model:** Road noise is diffuse rather than a single plane wave; a more accurate model would use a diffuse-field incidence.

(v) **No experimental validation:** Results have not been verified against in-situ measurements. A simple smartphone SPL measurement could provide a basic check of the predicted modal frequencies.

6 Conclusion

This study has presented a two-dimensional finite element model of a residential apartment, built from architectural drawing coordinates in COMSOL Multiphysics, to investigate the acoustic behaviour of the living room across 20 Hz to 500 Hz.

The key findings are:

(i) The irregular floor plan produces 19 acoustic room modes in the range 20 Hz to 96 Hz, with the dominant mode at 19.3 Hz corresponding to the longest dimension of the coupled apartment volume. The FSI coupling additionally reveals structural door resonances around 14 Hz.

(ii) Closing the living-room door raises the first living-room axial mode from 19.3 Hz to 22.9 Hz (+18.6%), while the full-apartment mode at 10.9 Hz is virtually unaffected (−0.1%), confirming it spans the entire building envelope regardless of door state.

(iii) Door-gap leakage of 1 mm to 5 mm provides essentially no low-frequency isolation; a proper door seal is required to achieve meaningful acoustic decoupling at bass frequencies.

(iv) Road-noise and loudspeaker sources excite the same room modes but with different spatial weightings, producing distinct SPL distributions that respond differently to treatment.

(v) Corner absorbers are most effective above 50 Hz; the lowest modes require DSP equalisation, multiple subwoofers with phase cancellation, or active noise control.

Practical recommendations: reposition the speakers away from the corner that most strongly excites the dominant mode; install corner bass traps at the two right-hand side corners of the living room (as modelled in Study 4); fit acoustic door seals to the living-room door; consider DSP equalization to address the residual low-frequency peaks.

Future work should include a full 3D model to capture vertical modes, measured impedance data for the wall and floor materials, experimental validation, and a diffuse-field road-noise model.

References

[1] Fahy, F., & Gardonio, P. (2007). *Sound and structural vibration* (2nd ed.). Academic Press.

[2] Kuttruff, H. (2009). *Room acoustics* (5th ed.). Spon Press.

[3] COMSOL Multiphysics. (2023). *Acoustics module user's guide*. COMSOL AB.

[4] Mechel, F. P. (2008). *Formulas of acoustics* (2nd ed.). Springer.

[5] ISO. (2003). *ISO 354:2003 — Acoustics: Measurement of sound absorption in a reverberation room*. International Organisation for Standardisation.

[6] Bies, D. A., Hansen, C. H., & Howard, C. Q. (2018). *Engineering noise control* (5th ed.). CRC Press.

A COMSOL Model Parameters

Table 8: Global model parameters as defined in COMSOL Global Definitions.

Parameter	Value	Unit	Description
`c_sound`	343	m/s	Speed of sound at 20 °C
`rho_air`	1.21	kg/m^3	Air density at 20 °C
`f_max`	500	Hz	Upper frequency bound
`h_max`	`c_sound/(6*f_max)`	m	Max mesh element size = 0.114 m
`alpha`	$0 \ldots 0.99$	—	Corner absorber absorption coefficient
`door_h`	0.8	m	Full door leaf height at `gap_h`=0
`door_w`	0.04	m	Door leaf width (wall thickness)
`gap_h`	$0.001 \ldots 0.005$	m	Door gap height (parametric sweep)
`E_wood`	1×10^9	Pa	Young's modulus, hollow-core door
`nu_wood`	0.3	—	Poisson's ratio, door leaf
`rho_wood`	500	kg/m^3	Density, door leaf
`eta_wood`	0.05	—	Structural loss factor, door leaf

B Polygon Coordinates

Table 9: Outer wall polygon vertices (44 points, `Yttre_med` layer). Origin at bottom-left of bounding box, Y-axis up. Scale factor: $\times\,(W_{\mathrm{real}}/10.5\,\mathrm{m})$.

Point	x (m)	y (m)
P1	3.1180	13.4445
P2	5.9802	13.4445
P3	5.9902	12.4515
P4	5.9902	12.1333
P5	6.0909	12.1333
P6	6.0909	12.4465
P7	8.0162	12.1512
P8	8.2965	13.1523
P9	10.1224	12.8840
P10	9.3156	7.2934
P11	6.0422	7.7668
P12	6.0422	7.9075
P13	5.9386	7.9075
P14	5.9386	7.1895
P15	6.0437	7.1895
P16	6.0437	7.6587
P17	9.3003	7.1880
P18	8.8200	3.8598
P19	6.0377	4.2562
P20	6.0377	6.3895
P21	5.9444	6.3895
P22	5.9444	6.1940
P23	5.7111	6.1940
P24	5.7111	6.0934
P25	5.9366	6.0934
P26	5.9366	4.2622
P27	4.4311	4.2622
P28	4.4311	6.0795
P29	5.2449	6.0795
P30	5.2449	6.1849
P31	4.4055	6.1849
P32	4.3845	8.8339
P33	4.3845	9.3486
P34	4.2771	9.3486
P35	4.2771	8.8109
P36	3.1048	8.8109
P37	3.1048	9.9812
P38	4.2712	9.9812
P39	4.2712	9.8727
P40	4.3672	9.8727
P41	4.3672	9.9808
P42	4.5432	9.9808
P43	4.5432	10.0829
P44	3.1180	10.0829

Table 10: Inner wall polygon vertices (10 points, `Innervägg` layer). Same coordinate system as outer wall.

Point	x (m)	y (m)
P1	6.0542	11.3333
P2	6.0542	8.7075
P3	5.9492	8.7075
P4	5.9492	8.9830
P5	5.8050	8.9830
P6	5.8050	9.9875
P7	5.3432	9.9875
P8	5.3432	10.0822
P9	5.9586	10.0822
P10	5.9586	11.3333

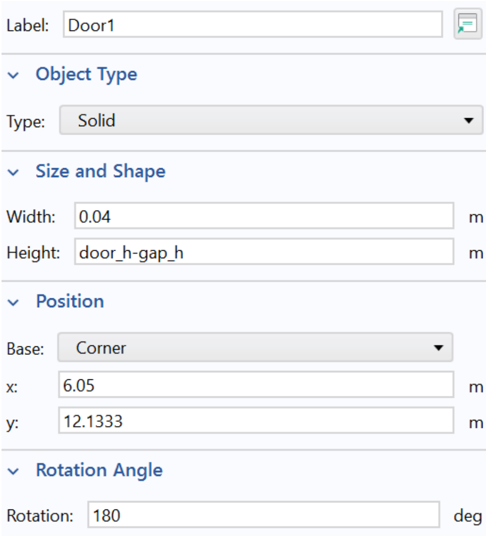
Table 11: Balcony polygon vertices (4 points, closed loop). Same coordinate system as outer wall.

Point	x (m)	y (m)
P1	6.0909	13.4445
P2	6.0909	12.5500
P3	7.9500	12.2500
P4	8.2000	13.1665

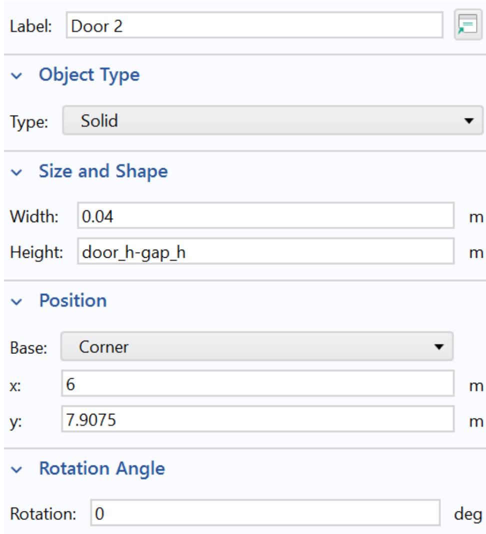
C Door Geometry and Base Corner Constraints

The screenshots in this appendix show the COMSOL geometry settings for each of the four door leaf rectangles in the model. All four doors share an identical setup: the height field contains the expression `door_h - gap_h`, the width equals the local wall thickness, and the base corner (bottom-left of the rectangle) is fixed so that the gap opens at the top of the frame as `gap_h` increases. The door leaf surfaces carry an impedance boundary condition ($Z_{\mathrm{door}} = 1.5 \times 10^4$ Pa s/m); the open gap strip above has no boundary condition assigned.

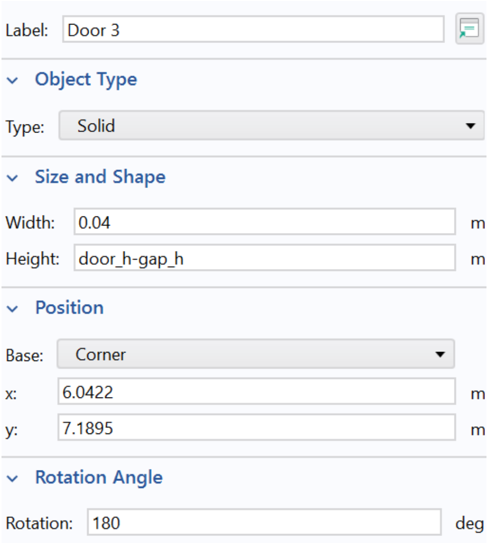
The only variations between the four doors are the **base corner coordinates** — which place each rectangle at its correct position within the apartment geometry — and the **rotation angle**, since some door frames are not axis-aligned due to the irregular floor plan. All other parameters (`door_h`, `gap_h`, Z_{door}, impedance expression) are identical and shared through the global parameter definitions.



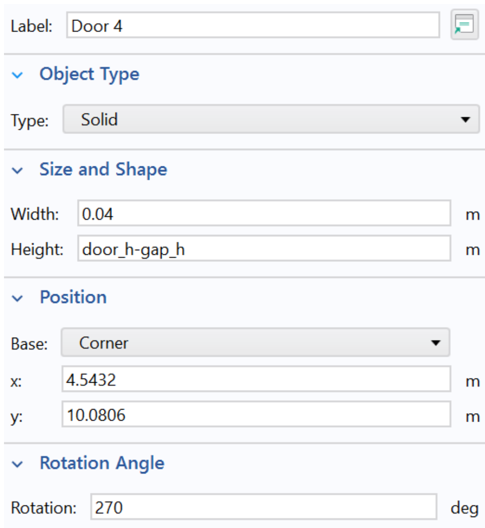
(a) Door 1



(b) Door 2



(c) Door 3



(d) Door 4

Figure 13: COMSOL settings for all four door leaf rectangles. Height = `door_h - gap_h`; only the base corner coordinates and rotation angle differ between doors.